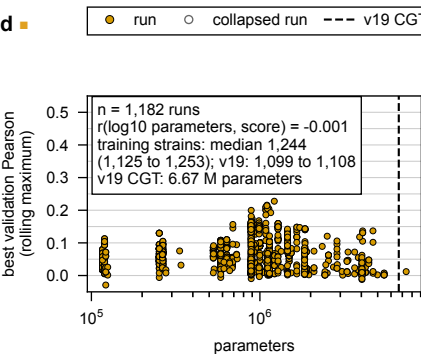
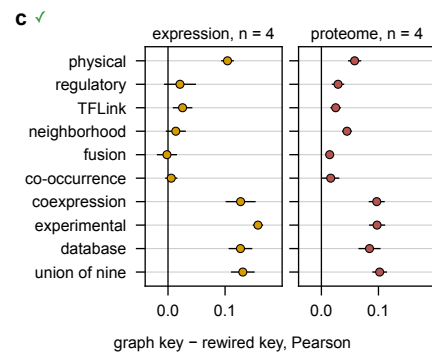
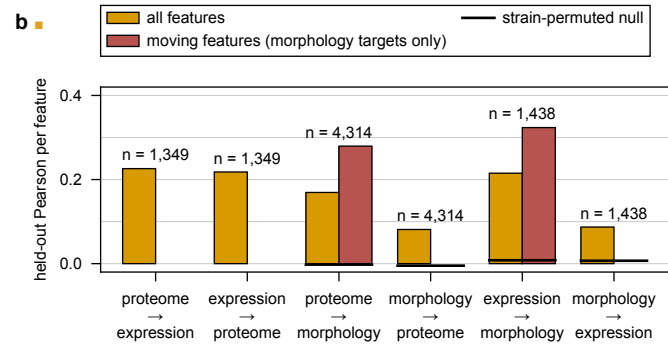
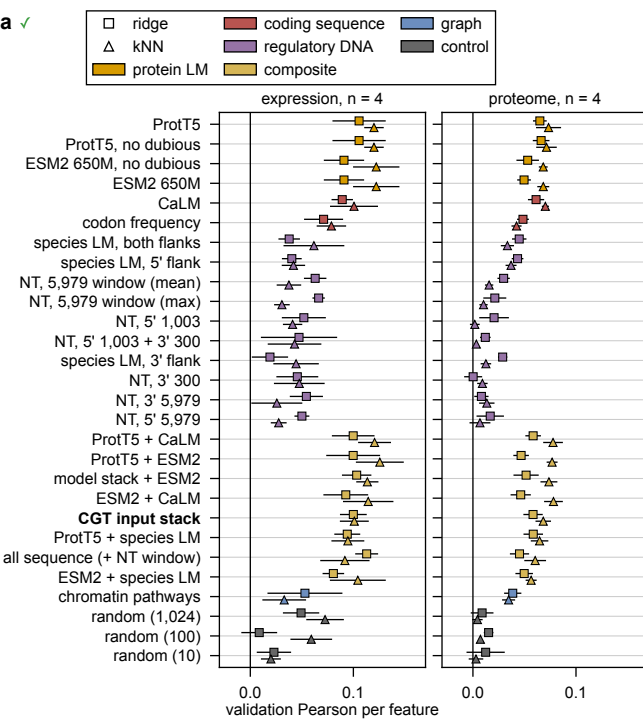




e ■ As drawn in Fig 1f

$$\beta_{i,t} = \sigma \left(\frac{(h_i W_Q) \cdot (\mathbf{p}_t W_K)}{\sqrt{d_k}} \right)$$

$$\Delta h_i = \sum_t \beta_{i,t} (\mathbf{p}_t W_V)$$

$$h_i^p = h_i + \Delta h_i$$

$$p = \{(e_t, \tau_t, m_t)\}_{t=1}^M$$

With one perturbed gene, weights that differ between genes need the function to act on each pair, a sigmoid. Methods write a softmax over t .

As run in v19 and v20 (null sink off)

$$\beta_{i,t} = \text{softmax}_{t \in [M]} \frac{(h_i W_Q) \cdot (\mathbf{p}_t W_K)}{\sqrt{d_k}}$$

$$\Delta h_i = \sum_t \beta_{i,t} (\mathbf{p}_t W_V)$$

$$h_i^p = h_i + \Delta h_i$$

$$p = \{(e_t, \tau_t, m_t)\}_{t=1}^M$$

Softmax over the perturbed genes: with one perturbed gene the weight is 1 for every gene and the update is the same vector for every gene.